In [25]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

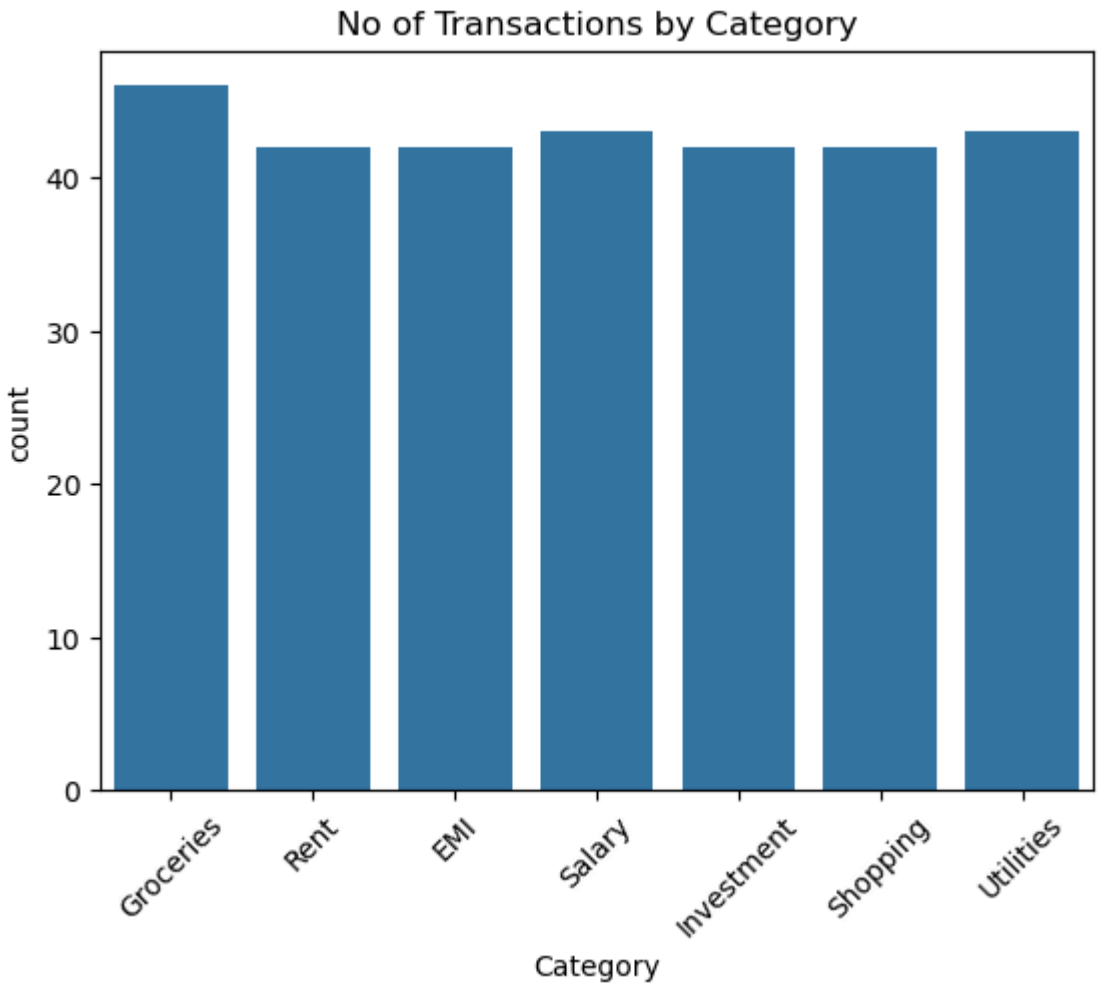
In [26]:
```
df = pd.read_csv("2. Finance_Analytics_Dataset.csv")
print(df)
```

```
     TransactionID        Date AccountType TransactionType   Category  Amount  \
0                1  01-01-2023     Savings           Credit  Groceries  120068
1                2  02-01-2023     Current            Debit       Rent  145359
2                3  03-01-2023      Credit            Debit        EMI  119216
3                4  04-01-2023     Savings            Debit        EMI  132038
4                5  05-01-2023     Current            Debit       Rent   75523
..             ...         ...         ...              ...        ...     ...
295            296  23-10-2023      Credit           Credit   Shopping  139266
296            297  24-10-2023      Credit            Debit       Rent  118849
297            298  25-10-2023     Savings            Debit  Groceries   84253
298            299  26-10-2023      Credit           Credit     Salary   67516
299            300  27-10-2023      Credit            Debit  Groceries  139612

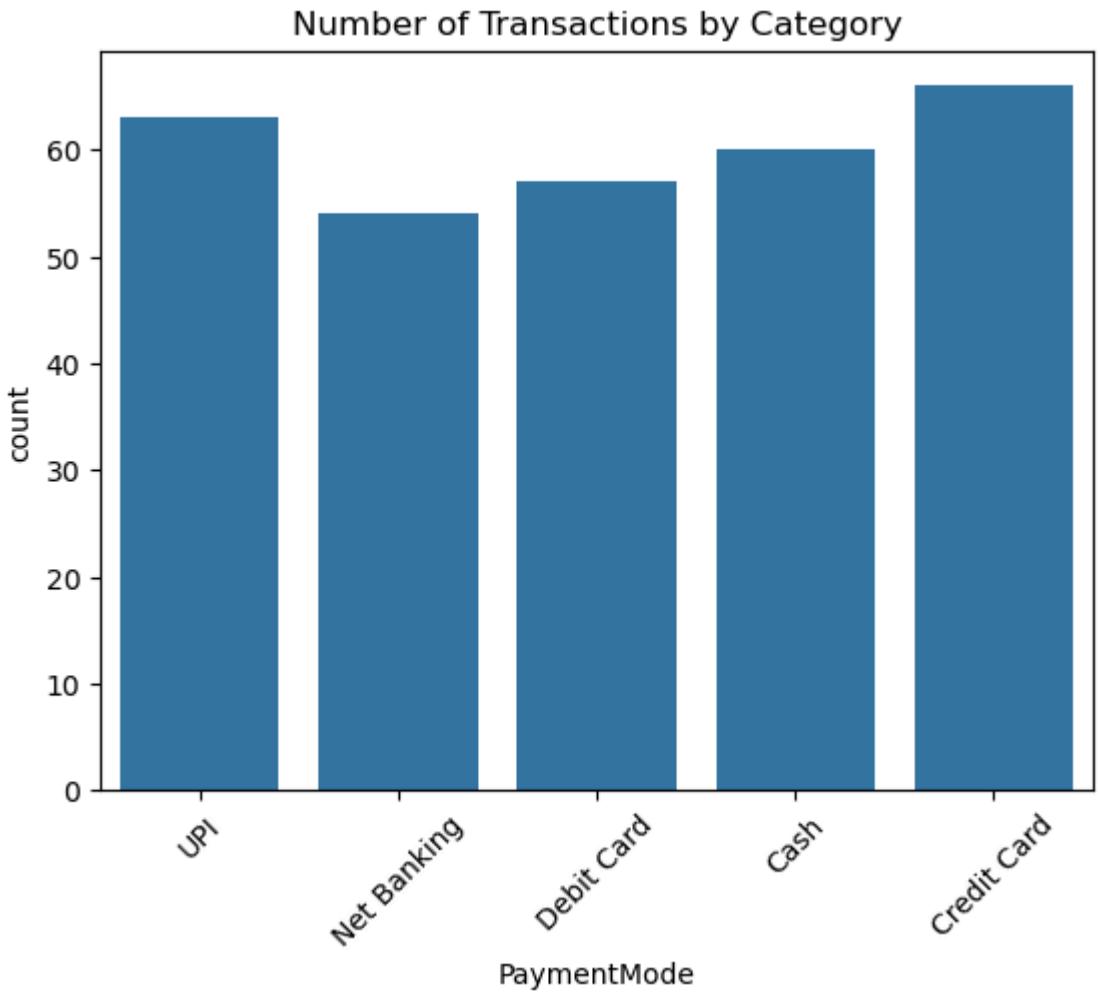
     Balance  PaymentMode CustomerSegment FraudFlag
0      16750         UPI          Retail       Yes
1     264540  Net Banking          Retail        No
2     100390         UPI       Corporate        No
3     199151   Debit Card             SME        No
4     400540        Cash             SME        No
..       ...         ...             ...       ...
295    50881   Debit Card       Corporate        No
296   303526         UPI          Retail        No
297   485578   Debit Card          Retail        No
298   299902         UPI          Retail        No
299   288776  Net Banking       Corporate        No

[300 rows x 10 columns]
```

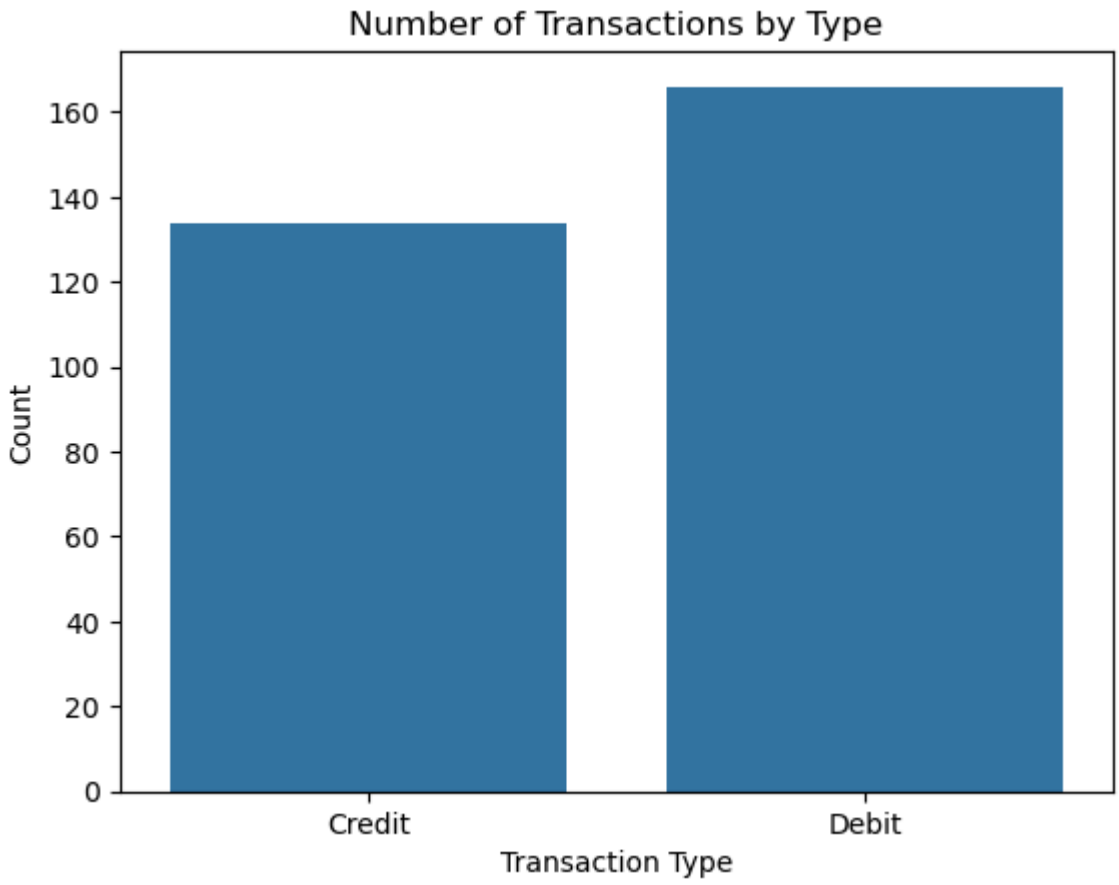
In [27]:
```
sns.countplot(data=df, x="Category")
plt.title("No of Transactions by Category")
plt.xticks(rotation=45)
plt.show()
```



In [28]:
```
sns.countplot(data=df, x="PaymentMode")
plt.title("Number of Transactions by Category")
plt.xticks(rotation=45)
plt.show()
```



In [29]:
```
sns.countplot(data=df, x="TransactionType")
plt.title("Number of Transactions by Type")
plt.xlabel("Transaction Type")
plt.ylabel("Count")
plt.show()
```



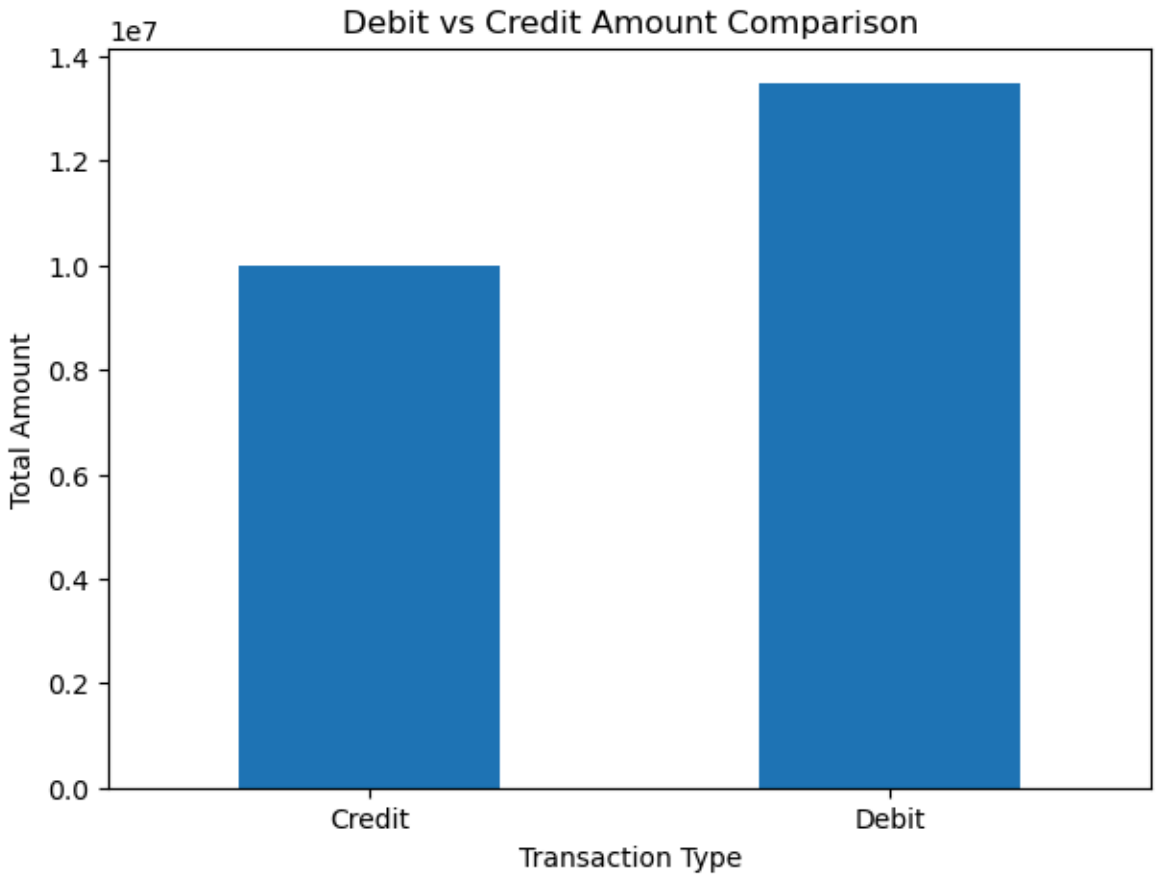
In [30]:
```
transaction_amount = df.groupby("TransactionType")["Amount"].sum()

plt.figure(figsize=(7, 5))

transaction_amount.plot(kind="bar")

plt.title("Debit vs Credit Amount Comparison")
plt.xlabel("Transaction Type")
plt.ylabel("Total Amount")

plt.xticks(rotation=0)
plt.show()
```



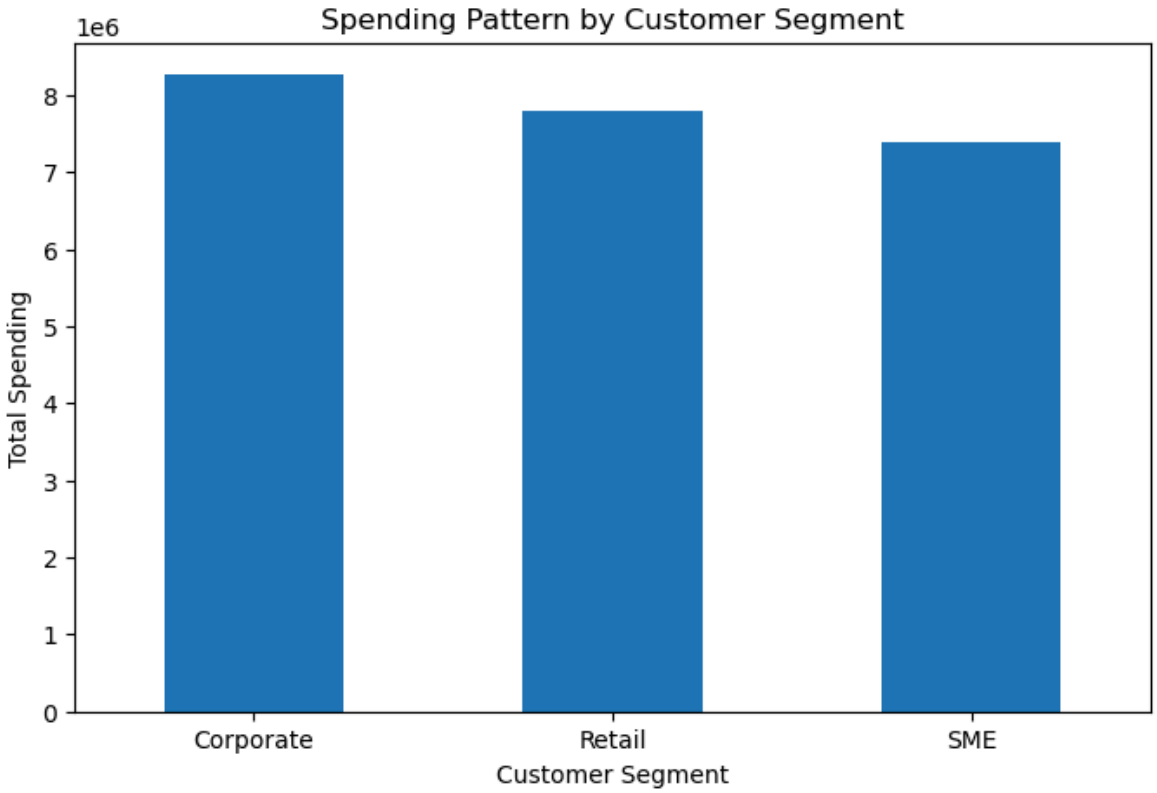
In [31]:
```
segment_amount = df.groupby("CustomerSegment")["Amount"].sum()

plt.figure(figsize=(8, 5))

segment_amount.plot(kind="bar")

plt.title("Spending Pattern by Customer Segment")
plt.xlabel("Customer Segment")
plt.ylabel("Total Spending")

plt.xticks(rotation=0)
plt.show()
```



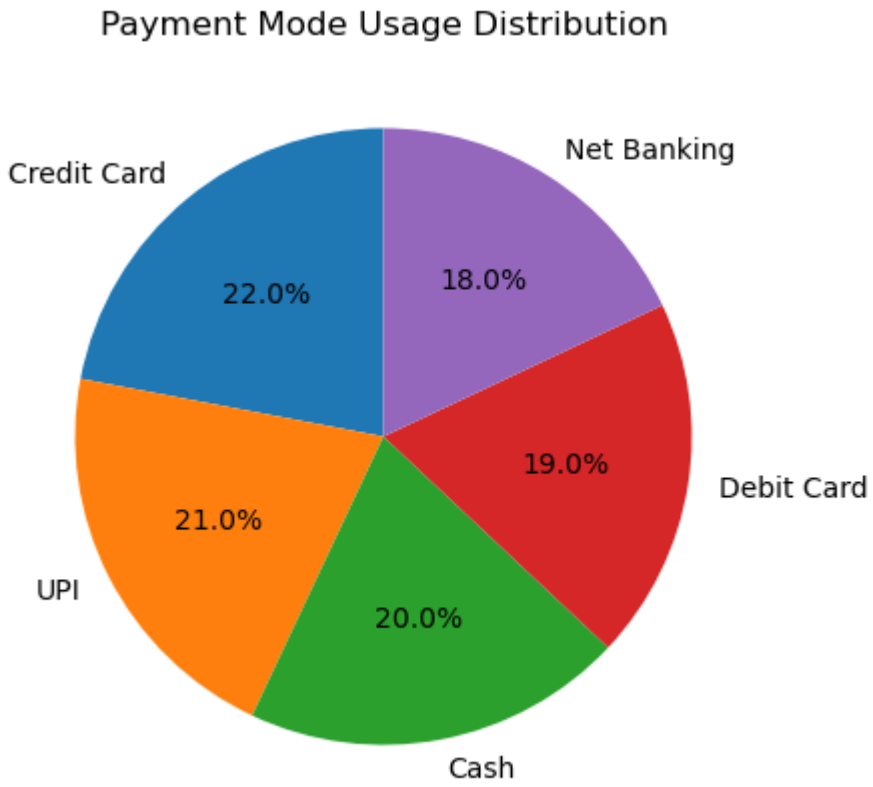
In [32]:
```
payment_counts = df["PaymentMode"].value_counts()

plt.figure(figsize=(5, 5))

plt.pie(
    payment_counts,
    labels=payment_counts.index,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Payment Mode Usage Distribution")

plt.show()
```



In [33]:
```
plt.figure(figsize=(10,6))
sns.boxplot(x="Category", y="Amount", data=df)

plt.title("Transaction Amount Distribution by Category")
plt.xlabel("Category")
plt.ylabel("Transaction Amount")
plt.xticks(rotation=45)
plt.show()
```

